MIL-C-21583(NOrd)

20 August 1958
SUPERSEDING
NAVORD OS 2676D
12 November 1954

MILITARY SPECIFICATION

CS MECEANISM Mark 1 MODS 3, 4, 9, AND 11

1. SCOPE

1.1 This specification establishes the requirements for the procurement of the CS Mechanism Mark 1 Mods 3, 4, 9, and 11. The CS Mechanism Mark 1 is a clock starter operated by hydrostatic pressure. It unlocks and starts an attached clock delay mechanism at the proper depth. The various mods differ only in minor variations.

2. APPLICABLE DOCUMENTS

2.1 The following specifications standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-P-116	Presentation, Methods of.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSC 1350

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DRAWING

BUREAU OF ORDNANCE

LD 130581	CS Mechanism Mark 1 Mod .
ID 130582	CS Mechanism Mark 1 MOD 2.
LD 139217	CS Mechanism Mark 1 Mod 9.
LD 169485	CS Mechanism Mark 1 Mod 11.
LD 291755	Test Pot Mark 3 Mod 1.
LD 295891	Container, Inner, for CS Mechanisms, Mark 1 type.
LD 295892	Container Outer, for C6 Mechanisms, Mark 1 type.
147619	Marking Instructions for Storage and Shipping Containers.

And all documents listed in the above LD's and drawings.

PUBLICATIONS

BUREAU OF ORDNANCE

NAVORD OP 400	General Instructions for the Design, Manufacture and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.).

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3. REQUIREMENTS.

3.1 General requirements.

3.1.1 The OS Mechanism Mark 1 shall be manufactured in accordance with the below listed LD's and all documents listed thereon.

Mod 3	LD 130581
Mod 4	LD 130582
Mod 9	LD 139217

3.1.2 The requirements and procedures established in CP 400 shall be complied with as applicable.

3.1.3 Preproduction sample. - Unless otherwise specified in the contract or order a preproduction sample shall be manufactured by the same procedures and processes and at the same location as proposed for full production. This sample will be tested for the purpose of determining compliance with the requirements of this specification, the drawings, and the contract. See 4.1.2.

3.1.4 Interchangeablity - Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict agreement with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the.fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings shall be made without prior approval of the Bureau of Ordnance.

3.2 Requests for deviation. - Requests for deviation from this specification, applicable drawings, specifications, publications, materials and processes specified herein, shall be as set forth in OP 400.

3.3 Conflicting requirements. - Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in Writing to the procuring agency or appointed agent for interpretation and clarification.

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3.4 Threads. - Unless otherwise specified all threads shall be in accordance with the the National Bureau of Standards Handbook H28 and supplements thereto. The class or fit for threads shall be as specified on the applicable drawings, and the NOT-GO gaging practice shall be as set forth in the supplement to the aforementioned Handbook H28.

3.5 Performance requirements and product characteristics The mechanism covered by this specification shall meet the following performance requirements and product characteris tics.

3.5.1 Flange Porositv. - The flange shall be capable of withstanding an air pressure of 60 pounds per square inch and shokr no leakage when tested as specified in 4.5.3.

3.5.2 Assembly. - The diaphram shall be assembled on the piston before assembly of the" diaphragm into the flange. The nut holding the diaphragm on the piston shall be tightened with a torque of from 25 to 35 pound-inches without distorting the diaphragm to assure that a watertight seal is made. The piston-diaphragm sub-assembly shall then be installed in the flange. The threads of the ring nut shall be coated with antiseize compound to prevent binding in the flange and special care shall be used to prevent the antiseize compound from contacting the diaphragm. The ring nut shall be tightened on the diaphragm while the piston is In the extended position. A torque of from 150 to 170 inch-pounds shall be used to tighten the ring nut without distorting the diaphragm, to assure that a watertight seal Is made.

3.5.3 Hand operation. - The mechanism shall be assembled with the safety device in the unlocked condition. The piston shall be depressed to its fully extended position end then allowed to return to its normal position There shall be no binding during the piston travel.

3.5.4 Leakage. - With the safety device in the unlocked condition. the assembled mechanism shall be capable of with-standing a pressure differential of 60 pounds per square inch, without leaking, when tested as specified in 4.5.6.

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3.5.5 Operating pressure. - The CS Mechanisms covered by this specification shall operate within the following pressure limits with no clock delay mechanism attached. Testing shall be as specified in 4.5.7.

(a) The piston shall move to its fully extended position when the pressure is Increased to some pressure between 2 and 6 pounds per square inch.

(b) The piston shall fully retract when the pressure is removed.

3.6 Workmanship and cleanliness. The mechanism shall be free from dust, dirt, metal filings, moisture, chips, and other foreign matter. Necessary precautions shall be taken during manufacture and assembly to insure these conditions.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS.

4.1 Government inspection. - The Government inspector will make such inspections as are necessary to determine that all components and assemblies are in accordance with the requirements of the contract, pertinent specifications and drawings. Unless otherwise specified, these inspections will be conducted in accordance with Instructions contained in OP 400.

4.1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination Of the acceptability of products submitted by a supplier for Government inspection will be in accordance with the provisions of MIL-STD-105.

4.1.2 Preproduction sample. - Unless otherwise specified for each contract or order, a representative sample of twelve (12) mechanisms shall be manufactured using the methods and procedures proposed for full production The Government Inspector shall perform, witness, or participate in such inspections and tests on the sample as are required for full production. Upon conditional acceptance by the inspector, the preproduction sample shall be forwarded, together with a report of inspection, to an activity designated in the contract or by the procuring agency, for the purpose of determining compliance with the requirements of the contract, specifications and drawings. Further production of the item by the

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contractor prior to approval of the contracting activity or the completion of the tests on preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order

4.1.2.1 Preproduction sample for a subsequent contract. - The necessity for a preproduction sample will be determined by the procuring agency when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory material as covered by this document.

4.1.3 Lot. - As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot" i.e. a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product In "inspection lots shall be as determined by the Government inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2 Contractor's inspection. - Contractort's inspection shall be conducted in accordance with provisions of "Inspection by Contractor" as contained in OP 400.

4.3 Classification. of tests. - The inspection and testing of mechanisms covered by this document shall be classified as follows:

(a) acceptance tests

(b) preproduction tests

4.3.l Acceptance tests. - Acceptance tests for acceptance of the mechanism covered by this specification shall consist of tests to determine compliance with all requirements specified in this document.

4.3.2 Peproduction tests. - Peproduction tests are those performed on samples submitted in accordance with 4.1.2 to determine that the lot and production meet all the requirements of this specification. Acceptance shall be based on no defects in a sample. Acceptable preproduction sample units shall not be released for service use and shall be stamped or marked as follows: "Sample - Not for Service Use".

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4.4 Test equipment

4.4.1 Test and inspection equipment and facilities. - Unless otherwise specified, the manufacturer shall furnish and maintain all necessary test equipment personnel.for performing all acceptance tests under the supervision of the Government inspector. The test equipment shall be adequate in quantity, and unless definite requirements are specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.1.1 Equipment required for performing these tests is as follows:

(a) Test Pot Mark 3 Mod 1 (BuOrd LD 291755) (Government will furnish).

(b) The contractor will furnish a test pot, and any other necessary equipment acceptable to the Government Inspector and the Bureau of Ordnance, to perform the flange porosity test as specified in paragraph 4.5.3.

(c) External air supply.

4.5 Tests.

4.5.1 Test conditions.- Unless otherwise specified, tests shall be performed at 80 and ambient relative humidity.

4.5.2 Visual inspection. - The mechanism and its component parts shall be inspected to verify that the materials, design, construction, physical dimensions and workmanship are in accordance with this document and the drawings

4.5.3 Flange porosity test. - To determine compliance with 3.5.1, the following test procedure shall apply. machining, and before anodizing, if the flange is aluminum alloy, the flange shall be mounted in a test pot (test item 4.4.1.1 (b)) with a banking plate and gasket sealing the diaphragm opening- Air at a pressure of 60 pounds per square inch shall be applied to one side of the flange for a period of at least 60 seconds. At least the last 30 seconds of this period shall be devoted to an examination of the exposed end

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of the flange under water for evidence of leakage. Flanges which show any leakage during this test shall be rejected.

4.5.4 Assembly. - The assembly procedure shall be checked to determine compliance with 3.5.2.

4.5.5 Hand operation. - The mechanism shall be tested to determine compliance with 3.5.3. If any binding occurs during this test the mechanism shall be rejected.

4.5.6 Leakage test. - To determine compliance with 3.5.4 the leakage test shall be conducted with the Test Pot Mark 3 Mod 1, (test item 4.4.1.1 (a)) following the test procedure shown on Bureau of Ordnance Drawing No. 1362576.

4.5.7 Operating pressure test. - After the leakage test, the leakage test fixture on the Test Pot (test item 4.4.1.1 (a)) shall be replaced by the actuation test fixture and the following tests shall be made:

(a) The pressure shall be increased slowly to some pressure within the range of operating pressures required by 3.5.5 If the piston does not move to its fully extended position within the pressure limits required by 3.5.5, the mechanism shall be rejected.

(b) The pressure shall then be decreased slowly If the piston does not move to its fully retracted position when the pressure has been removed, the mechanism shall be rejected.

Each mechanism being tested, shall be operated as described above at least 3 times before acceptance and a record shall be kept of the average operating pressure of each mechanism.

4.5.8 Acceptance criteria.- Failure of any mechanism to meet any of the requirements and tests of this specification shall result in rejection of the mechanism. If the number of failing mechanisms exceeds the number permitted by the associated criteria, the lot represented will be rejected.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning and drying. - Unless otherwise specified, Clock Starter Mechanisms shall be cleaned in accordance with Method C-1 of Military Specification MIL-P-116 and dried in accordance with one or more of the procedures given in Military Specification MIL-P-116, provided the selected procedure IS not injurious to the item.

5.1.1.2 Preservative application. - (Not applicable).

5.1.1.3 Unit packing. - Each Clock Starter Mechanism shall be individually packaged in accordance with Bureau of Ordnance LD 295891.

5.1.2 Level B. - (Not applicable).

5.1.3 Level C. - (Not applicable

5.2 Packing

5.2.1 Level A.

5.2.1.1 Exterior containers. - Four (4) Clock Starter Mechanisms shall be packed in accordance with Bureau of Ordnance LD 295892.

5.2.2 Level B. - (Not applicable).

5.2.3 Level C. - (Not applicable).

5.3 Marking.

5.3.1 Normal markings. - In addition to any special marking required by the contract or order, all markings shall be in accordance with Military Standard MIL-STD-129.

5.3.1.1 For detailed marking requirements, see Bureau of Ordnance Drawing Number 1476194

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6. NOTES

6.1 Ordering data. - Procurement documents should specify the title, number and date of this specification, and exceptions to this specification applicable drawings and other documents. Procurement documents shall specify Level A packaging and packing requirements for all shipments.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever: and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, Or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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